To some extent this has been safeguarded by slugging the remaining supplies relative to -5V. In fact it is essential that VBB is never more +ve than VSS If independant supplies are used, then the -12V supply <u>must</u> be switched on first and off last.

Bias failure

If the -5V supply fails the bleed resistor will keep the bias voltage close to ground but the safest approach is to use the -5V supply to switch the +5V and +12V supplies. To help with this approach the -5V supply has been taken out to an edge connector. Pin 73

Testing

Now observing the usual precautions when handling MOS insert just <u>one</u> 4116 in position X23 (the right way round).

Using MIKBUG or equivalent insert the following sequence of words at one or a number of addresses in the range 0000 to 3FFF.

INPUT 0000,0000

RESULT 0XXX,XXXX   'X' = don't care state.

INPUT 1111,1111

RESULT 1XXX,XXXX

Move the device from X23 to X24 then

INPUT 0000,0000

RESULT X0XX,XXXX

INSERT 1111,1111

RESULT X1XX,XXXX

Then move the device from X24 to X25 and so on until the complete byte has been checked. Now change the address selection switch X10 so that 4 is on and continue as above with the second block of memory. (4000-7FFF).

Once this has all checked out insert all the RAM chips and run the memory test programme attached.

CONSTRUCTION

Connections and Links

Initially, do not connect to the additional decode inputs (X17 pins 2-6) or the write protect input (X18 pin 9).

Make all wire links, the holes are large enough to enable through links to be made by looping in and out of each hole.

Insert links 1 and 2.

Insert the transistor, resistors and capacitors and all RAM sockets (X23 - X38). Use a clean, small soldering iron bit and minimum amount of solder.

Examine the PCB carefully, with a magnifying glass if possible for solder bridges - most of your problems will be of this nature.

Check for power supply shorts.

Back Wiring

If a back plane is used isolate the clock outputs and the refresh control signals. Otherwise the connections are directly compatible with 77-68.

Insert the board and power up. Check for +12V, -5V and +5V at each RAM position - do not insert any I.C.'s yet.

Set the DIP switch (X10) to 1 and 4. This will position the RAM block (X23 - X30) in the address space 0000-3FFF (HEX) and the remaining block at 4000-7FFF.

Fill all I.C. positions except RAM, then power up.

Check that the counter chain is working by measuring approximately 2V on X8 pin 8 and X16 pin 12. On X22 pin 9 measure 0.25 - 0.5 volts.

Do not proceed further until all these conditions have been satisfied.

Inserting the 4116's

The substrate bias for these devices is supplied by the -5V rail and it is essential that no other pin on the device is ever subjected to a voltage more negative than this, that includes the period when the power supplies are switched on.